

Prototype IRS-Based Nonprofit Directory with MongoDB Search

Overview

Build a small prototype for an open nonprofit data directory using publicly available U.S. IRS nonprofit data.

This is **Phase 0** of a larger organization intelligence directory for Chambey. The long-term direction is conceptually similar to a lightweight Duns & Bradstreet-style business/organization directory, but this first task should stay small, concrete, and focused.

The goal is not to copy D&B, GuideStar, or Charity Navigator. Those are only reference models for future inspiration.

For this phase, the goal is to prove that we can collect, structure, store, and search legitimate public nonprofit data in a clean and compliant way.

Primary Objective

Create a minimal, structured, searchable directory of U.S. tax-exempt organizations using open IRS data.

The prototype should:

- Identify available IRS nonprofit datasets
 - Use official public data sources where possible
 - Avoid unnecessary scraping
 - Ingest IRS data into MongoDB
 - Preserve both raw and cleaned data
 - Normalize core organization fields
 - Make records searchable
 - Document the technical approach clearly
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Required Starting Source

Start with IRS Tax Exempt Organization Search data.

Prioritize official bulk downloads over scraping the IRS search website.

Investigate these IRS sources first:

- IRS Publication 78 data
- IRS Exempt Organizations Business Master File Extract
- IRS Auto-Revocation List

- IRS Form 990 data, only if appropriate for later expansion
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Reference Models

Use these only for inspiration:

- D&B Business Directory
- GuideStar nonprofit search
- Charity Navigator charity discovery

Do **not** attempt to replicate their full feature sets in this phase.

Additional Research Sources

You may also explore:

- Nonprofit Open Data Collective
 - Data.gov nonprofit-related datasets
 - Open-source intelligence concepts related to:
 - Public data discovery
 - Data enrichment
 - Entity validation
 - Entity linkage
 - Data quality checks
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Scope of Work

1. Data Discovery

Identify what IRS nonprofit datasets are available.

For each dataset, document:

- Dataset name
 - What it contains
 - File format
 - Access method
 - Update frequency, if available
 - Coverage and limitations
 - Whether it should be used in Phase 0 or saved for later
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2. Access and Compliance Review

Before building the pipeline, document the cleanest and most compliant way to collect the data.

Include:

- Whether bulk download is available
- Whether scraping is necessary or avoidable
- Any terms of use concerns
- Any robots.txt or rate-limit concerns if web access is involved
- Attribution requirements
- Risks or limitations

Important:

Do not scrape first. Investigate official bulk downloads first.

Prefer official downloadable files over scraping public web pages.

3. Ingestion Prototype

Build a small repeatable ingestion pipeline.

The pipeline should:

- Download or load IRS data
- Store the untouched raw file or raw records
- Parse the data
- Convert records into a clean normalized format
- Save cleaned records into MongoDB
- Log basic ingestion results

The pipeline does not need to be production-grade yet, but it should be runnable and understandable.

4. Data Cleaning and Normalization

Normalize the following fields where available:

- Organization name
- EIN
- Tax-exempt status
- Revocation status, if available
- Address
- City
- State

- ZIP code
- Country, if available
- NTEE code/category, if available
- Subsection / exemption type, if available
- Ruling date, if available
- Deductibility status, if available
- Source dataset
- Last updated date
- Raw source reference

Handle duplicates and missing values carefully.

Do not invent missing information.

5. Storage Design

Create a storage-agnostic data model.

At minimum, propose these layers:

Raw Source Layer

Stores original downloaded files or raw records.

Canonical Organization Layer

Stores one clean organization profile per EIN where possible.

Search Index Layer

Supports searching by:

- Organization name
- EIN
- City
- State
- ZIP code
- Category
- Status

The prototype should use MongoDB for searchable records, but the data model should be explained in a way that could later move to another database if needed.

6. Minimal Search Experience

Create a simple way to search records.

This can be one of the following:

- Basic API endpoint
- Simple command-line search
- Minimal admin page
- MongoDB query examples

Search should support at least:

- Organization name
 - EIN
 - State
 - City
 - Status
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Deliverables

1. Working Prototype

A runnable ingestion pipeline that loads IRS nonprofit data into MongoDB.

2. Sample Dataset

A small persisted dataset with sample records.

3. Data Model Proposal

Include:

- Entity schema
- Field definitions
- Raw vs. cleaned data explanation
- Expansion notes

4. Technical Brief

Include:

- Data sources used
- Why those sources were selected
- Access method
- Compliance considerations

- Pipeline steps
- Data schema
- Known limitations
- Recommended next steps

The brief can be delivered as:

- Markdown
- PDF
- ODF
- Repository README

5. Demo

At completion, walk through:

- How the data was accessed
- How the pipeline works
- What the MongoDB records look like
- How search works
- What should be improved next

Definition of Done

This issue is complete when:

- IRS nonprofit data has been successfully ingested
- Records are searchable in MongoDB
- Raw and cleaned data are both preserved
- A clear schema exists
- A short technical brief is completed
- The approach is compliant and repeatable
- A feasible path for scaling is identified

Out of Scope for Phase 0

Do not build:

- Ratings or scores
- Paid data integrations
- User accounts
- Authentication
- Litany integration
- Full business intelligence profiles

- A full D&B clone
- A complex UI

Do not use:

- Proprietary data
 - Restricted data
 - Private data
 - Unverified scraped data as the primary source
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Success Criteria

This task is not only about whether the prototype works.

It is also meant to show:

- How public data is evaluated
 - How compliance is handled
 - How cleanly information is structured
 - How well the work is documented
 - Whether this can become a foundation for Chambey's future organization intelligence directory
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Preferred Technical Approach

- Favor clarity over speed
 - Use widely adopted open-source tools
 - Keep the system simple
 - Make the pipeline reproducible
 - Explain tradeoffs clearly
 - Preserve raw data before cleaning
 - Avoid overbuilding the UI
 - Focus on clean data foundations first
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Suggested Tech Stack

The implementation stack is flexible, but a simple option would be:

- Python or Node.js for ingestion
- MongoDB for storage and search
- CSV/JSON parsing tools
- README documentation
- Optional: simple API using FastAPI, Express, or similar

The final choice is up to the developer, but the reason for the choice should be documented.

Notes

This is a small foundation-building task.

The long-term direction is an organization intelligence directory that may eventually connect with Litany and Chambey's broader data platform. For now, keep the work narrow:

IRS nonprofit data → clean ingestion → MongoDB search → documented schema → demo.